Volume & 3D Shapes

Year 4 · **Homework Worksheet**

Worked Example

Worked example: What is the volume of a cuboid measuring 5cm × 2cm × 2cm?

Answer: Volume = 5 × 2 × 2 = 20 cm^3.

Easy / Standard (15 questions)

1. What is the volume of a cuboid measuring 4cm × 2cm × 2cm?
2. How many faces does a cuboid have?
3. Which 3D shape has 5 faces, 9 edges and 6 vertices? A) cube B) triangular prism C) cone D) sphere
4. How many edges does a cuboid have?
5. How many vertices does a cuboid have?
6. What is the volume of a cuboid measuring 5cm × 2cm × 1cm?
7. Which shape has exactly ONE flat face and one curved surface, coming to a point? A) sphere B) cylinder C) cone D) cube
8. What is the volume of a cube with sides of 3cm?
9. Select ALL of these 3D shapes that have at least one flat (straight- edged) face: A) sphere B) cube C) cylinder D) cuboid E) cone
10. How many faces does a cube have?
11. What is the volume of a cuboid measuring 6cm × 1cm × 2cm?
12. Which shape has 6 identical square faces? A) cuboid B) cube C) cylinder D) triangular prism
13. How many edges does a cube have?
14. What is the volume of a cuboid measuring 7cm × 1cm × 1cm?
15. How many vertices does a cube have?

Standard (15 questions)

1. A cuboid measures 3cm × 3cm × 2cm. What is its volume?
2. Which cuboid has a volume of 20cm^3? A) 5×2×2 B) 4×5×1 C) both A and B D) neither
3. A cuboid measures 6cm × 2cm × 1cm. What is its volume?
4. How many faces does a square- based pyramid have?
5. How many vertices does a cone have?
6. Which 3D shape has 2 triangular faces and 3 rectangular faces? A) cube B) triangular prism C) cone D) cylinder
7. A cube has a volume of 27cm^3. What is the length of one side?
8. Select ALL of these cuboids that have a volume of MORE than 15cm^3: A) 4×2×2 B) 3×3×2 C) 5×1×2 D) 2×2×2 E) 4×1×4
9. How many edges does a triangular prism have?

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10. How many faces does a cylinder have (flat and curved together)?

11. A cuboid measures 4cm × 4cm × 1cm. What is its volume?

12. Which shape has 6 faces, 12 edges and 8 vertices, with all faces equal squares? A) cuboid B) cube C) triangular prism D) cone

13. A cube has sides of 4cm. What is its volume?

14. How many vertices does a triangular prism have?

15. A cuboid measures 3cm × 2cm × 5cm. What is its volume?

Standard / Advanced (15 questions)

1. A cuboid's height works out to twice its width. Given a width of 2cm and a length of 3cm, calculate the volume.

2. Take an edge length of 2cm and add 1cm more — that is the edge of a cube. What volume does this cube have? A) $8cm^3$ B) $27cm^3$ C) $9cm^3$ D) $6cm^3$

3. Matching the volume of a 2cm-sided cube exactly, a certain cuboid has a length of 4cm and a width of 1cm. What must its height be?

4. The volume of a cube is the same as a cuboid measuring 4cm×2cm×1cm. What is the length of the cube's edge?

5. A tank measures 8cm × 5cm × 2cm and is half full of water. What volume of water is in the tank?

6. A cube-shaped box has a volume of $64cm^3$. What is the length of one edge, and how many faces does it have?

7. A cuboid box has a volume of $48cm^3$. Its length is 6cm and width is 4cm. What is its height? A) 2cm B) 3cm C) 4cm D) 5cm

8. Three identical 2cm cubes are stacked directly on top of each other. What is the total volume, and the height, of the stack?

9. Select ALL of these boxes below that could be exactly filled by 24 unit cubes with no gaps: A) 4×3×2 B) 6×2×2 C) 8×3×1 D) 4×2×2 E) 2×2×6

10. A cuboid measures 6cm × 4cm × 2cm. It is cut exactly in half, dividing its 2cm height into two equal layers. What are the dimensions of each half?

11. A cuboid has a volume of $40cm^3$. Its length is twice its width, and its height is 5cm. What are its length and width?

12. A container measuring 8cm×5cm×3cm is completely filled with water, then poured into empty bottles that each hold $20cm^3$. How many bottles can be completely filled, and how much water is left over?

13. A fish tank measuring 10cm×5cm×4cm is half-filled with water, then a further $50cm^3$ of water is added. What is the total volume of water in the tank now?

14. A cuboid box is made from edges of 4cm, 3cm and 1cm. What is the volume of the box?

15. A cuboid's length, width and height are 6cm, 4cm and 2cm. If the height is doubled, what is the new volume? A) $72cm^3$ B) $84cm^3$ C) $96cm^3$ D) $108cm^3$

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